

Updated Vision

HormoneBench AI is an AI-powered ecosystem advancing women's hormonal health through research, education, clinical decision support, and public awareness.

Instead of serving only researchers, it serves **four primary audiences**.

HormoneBench AI Ecosystem

1. Research Hub (Researchers & Universities)

The flagship platform for scientific research.

Features

- Multi-dataset upload (CSV, Excel)
 - AI Dataset Validator
 - Dataset Cards
 - Model Cards
 - Benchmark Leaderboards
 - Foundation Models
 - Explainable AI
 - Research Copilot
 - Biomedical Knowledge Graph
 - Literature Review Assistant
 - Statistical Analysis
 - Research Report Generator
 - Open API
 - Collaboration Workspace
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2. Student Academy (Students & Learners)

A complete learning portal for students interested in AI, medicine, biology, and public health.

Learning Paths

- Women's Hormonal Health 101
- Endocrinology Basics
- PCOS
- Thyroid Disorders
- Menstrual Cycle Science
- Menopause
- Fertility
- Nutrition & Hormones
- AI in Healthcare
- Biomedical Data Science
- Machine Learning for Healthcare
- Research Methodology
- Scientific Writing

AI Tutor

Students can ask:

Explain estrogen like I'm 15.

What causes insulin resistance?

Difference between PCOS and Endometriosis?

Explain LH/FSH ratio.

Gemini provides interactive explanations.

3. Patient Awareness Center

This is **education only**, not diagnosis.

Include a clear disclaimer:

"HormoneBench AI provides educational information and is not a substitute for professional medical advice, diagnosis, or treatment."

Educational Modules

- Understanding Hormones
- Menstrual Health
- PCOS Awareness
- Endometriosis
- Menopause
- Fertility Education
- Thyroid Health
- Nutrition
- Mental Health & Hormones
- Lifestyle

Each topic includes:

- Simple explanations
 - Visual diagrams
 - FAQs
 - Myth vs Fact
 - When to seek medical care
 - Trusted resources
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4. Healthcare Professional Portal

Designed for clinicians, nutritionists, and researchers.

Features

- Clinical Guidelines
 - Research Summaries
 - Biomarker References
 - Evidence Explorer
 - Literature Updates
 - Dataset Repository
 - AI Validation Tools
 - Collaboration
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AI Agents

Research Agent

For scientists.

Capabilities:

- Analyze datasets
 - Recommend preprocessing
 - Benchmark models
 - Generate reports
 - Find literature
 - Explain predictions
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Student Mentor Agent

For students.

Can:

- Teach concepts
 - Quiz users
 - Generate study notes
 - Explain diseases
 - Create flashcards
 - Recommend learning resources
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Hormone Education Agent

For the public.

Can answer questions like:

- Why is my cycle irregular?
- What is estrogen?
- What foods affect hormones?
- How does stress influence hormones?

Responses should emphasize education and encourage consultation with healthcare professionals when appropriate.

Nutrition Agent

Built with dietitian-reviewed content.

Can:

- Explain balanced diets
- Discuss nutrients involved in hormonal health
- Share general healthy eating guidance
- Suggest meal planning principles

Should avoid personalized medical nutrition therapy without appropriate clinical oversight.

Research Writing Agent

Supports researchers by helping draft:

- Literature reviews
 - Abstracts
 - Methods
 - Discussion sections
 - Research proposals
 - Grant applications
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Data Scientist Agent

Can:

- Detect missing values
 - Identify outliers
 - Suggest preprocessing
 - Recommend models
 - Compare algorithms
 - Evaluate fairness
 - Generate Python code
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New Modules

Awareness Center

Daily articles

Weekly awareness campaigns

Hormone Myth Busters

Expert interviews

Infographics

Research highlights

Community

Forums

Student groups

Research groups

Women in STEM

University chapters

Events

Resource Library

Research papers

Clinical guidelines

Open datasets

Books

Courses

Podcasts

Videos

Downloadable resources

Research Challenges

Monthly AI competitions

Dataset challenges

Leaderboard

Student competitions

Hackathons

Publications

Research outputs

Preprints

Case studies

White papers

Conference papers

Expert Network

Microbiologists

Endocrinologists

Gynecologists

Nutritionists

Epidemiologists

AI researchers

Data scientists

Updated Homepage Navigation

- Home
 - Research
 - AI Agents
 - Learn
 - Awareness
 - Resources
 - Community
 - Publications
 - About
 - Contact
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Updated User Journey

Researchers

→ Upload datasets → AI validation → Benchmark → Reports → Publish

Students

→ Learn → AI Tutor → Quizzes → Certificates → Projects

Patients

→ Awareness → Educational AI → Lifestyle Guidance → Resources → Professional Care Information

Healthcare Professionals

→ Evidence → Guidelines → Research → Collaboration

Gemini-Powered Features for the Hackathon

- AI Dataset Understanding
 - Multi-file CSV Upload & Automatic Schema Matching
 - Literature Summarization
 - Research Copilot
 - Student Tutor
 - Hormone Education Assistant
 - Research Report Generator
 - Statistical Analysis Assistant
 - Biomedical Knowledge Graph Q&A
 - Multilingual educational support
 - Voice-based educational conversations (future roadmap)
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Long-Term Vision

Position HormoneBench AI as **the world's first open AI ecosystem dedicated to women's hormonal health**, where researchers accelerate discovery, students build expertise, healthcare professionals access evidence-based resources, and the public learns through trusted, AI-assisted education.

This broader vision strengthens the project for hackathons and innovation programs because it demonstrates impact across **research, education, healthcare, and public awareness**, while keeping the research platform as the core innovation.